

STEP 1 — Install and Import Required Libraries

```
# Install required libraries
!pip install transformers
!pip install torch
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.6.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->tr
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->tr
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggi
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0-
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0-
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggi
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hug
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch) (3.0.3)
```

```
# Import required libraries
from transformers import pipeline
import torch

print("Libraries imported successfully")

Libraries imported successfully
```

STEP 2 — Load Translation Model

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# English to French translator
model_name_en_fr = "Helsinki-NLP/opus-mt-en-fr"
tokenizer_en_fr = AutoTokenizer.from_pretrained(model_name_en_fr)
model_en_fr = AutoModelForSeq2SeqLM.from_pretrained(model_name_en_fr)

def translator_en_fr(text):
    inputs = tokenizer_en_fr(text, return_tensors="pt", padding=True, truncation=True)
    translated_tokens = model_en_fr.generate(**inputs, max_length=512)
    return [{'translation_text': tokenizer_en_fr.decode(translated_tokens[0], skip_special_tokens=True)}]

# English to German translator
model_name_en_de = "Helsinki-NLP/opus-mt-en-de"
tokenizer_en_de = AutoTokenizer.from_pretrained(model_name_en_de)
model_en_de = AutoModelForSeq2SeqLM.from_pretrained(model_name_en_de)

def translator_en_de(text):
    inputs = tokenizer_en_de(text, return_tensors="pt", padding=True, truncation=True)
    translated_tokens = model_en_de.generate(**inputs, max_length=512)
```

```
return [{'translation_text': tokenizer_en_de.decode(translated_tokens[0], skip_special_tokens=True)}]

# English to Hindi translator
model_name_en_hi = "Helsinki-NLP/opus-mt-en-hi"
tokenizer_en_hi = AutoTokenizer.from_pretrained(model_name_en_hi)
model_en_hi = AutoModelForSeq2SeqLM.from_pretrained(model_name_en_hi)

def translator_en_hi(text):
    inputs = tokenizer_en_hi(text, return_tensors="pt", padding=True, truncation=True)
    translated_tokens = model_en_hi.generate(**inputs, max_length=512)
    return [{'translation_text': tokenizer_en_hi.decode(translated_tokens[0], skip_special_tokens=True)}]

print("Translation models loaded successfully")
```

tokenizer_config.json: 100% 42.0/42.0 [00:00<00:00, 2.99kB/s]

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
source.spm: 100% 778k/778k [00:00<00:00, 13.3MB/s]

target.spm: 100% 802k/802k [00:00<00:00, 44.3MB/s]

vocab.json: 1.34M/? [00:00<00:00, 40.7MB/s]

/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip ins
warnings.warn("Recommended: pip install sacremoses.")

pytorch_model.bin: 100% 301M/301M [00:04<00:00, 136MB/s]

Loading weights: 100% 258/258 [00:00<00:00, 705.31it/s, Materializing param=model.shared.weight]

model.safetensors: 100% 301M/301M [00:05<00:00, 123MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, bu
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, bu
generation_config.json: 100% 293/293 [00:00<00:00, 23.9kB/s]

config.json: 1.33k/? [00:00<00:00, 64.1kB/s]

tokenizer_config.json: 100% 42.0/42.0 [00:00<00:00, 2.64kB/s]

source.spm: 100% 768k/768k [00:00<00:00, 35.8MB/s]

target.spm: 100% 797k/797k [00:00<00:00, 35.9MB/s]

vocab.json: 1.27M/? [00:00<00:00, 24.6MB/s]

pytorch_model.bin: 100% 298M/298M [00:05<00:00, 67.8MB/s]

Loading weights: 100% 258/258 [00:00<00:00, 512.34it/s, Materializing param=model.shared.weight]

model.safetensors: 100% 298M/298M [00:04<00:00, 103MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, bu
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, bu
generation_config.json: 100% 293/293 [00:00<00:00, 26.4kB/s]

config.json: 1.39k/? [00:00<00:00, 106kB/s]

tokenizer_config.json: 100% 44.0/44.0 [00:00<00:00, 4.47kB/s]

source.spm: 100% 812k/812k [00:00<00:00, 9.85MB/s]

target.spm: 100% 1.07M/1.07M [00:00<00:00, 36.2MB/s]

vocab.json: 2.10M/? [00:00<00:00, 20.3MB/s]

pytorch_model.bin: 100% 306M/306M [00:09<00:00, 34.7MB/s]

Loading weights: 100% 258/258 [00:00<00:00, 617.51it/s, Materializing param=model.shared.weight]

model.safetensors: 100% 306M/306M [00:05<00:00, 72.7MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, bu
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, bu
generation_config.json: 100% 293/293 [00:00<00:00, 14.1kB/s]

Translation models loaded successfully

STEP 3 — Prepare Input Sentences

```
sentences = [  
    "Artificial intelligence is transforming modern technology.",  
    "The weather is very pleasant today.",  
    "Machine learning models require large amounts of data.",
```

```

    "How are you doing today?",
    "The meeting will start at 10 AM tomorrow.",
    "Researchers are developing new methods for climate prediction.",
    "Social media influences people's opinions.",
    "She completed her assignment before the deadline.",
    "The internet connects millions of devices worldwide.",
    "Education plays an important role in society.",
    "Natural language processing enables computers to understand human language.",
    "Can you help me solve this problem?",
    "Technology improves communication across the world.",
    "Students are learning machine translation in this laboratory.",
    "The government announced a new economic policy."
]

```

STEP 4 — Perform Translation

```

print("English to French Translation\n")

for sentence in sentences:
    result = translator_en_fr(sentence)
    print("English:", sentence)
    print("French :", result[0]['translation_text'])
    print()

```

English to French Translation

English: Artificial intelligence is transforming modern technology.
 French : L'intelligence artificielle transforme la technologie moderne.

English: The weather is very pleasant today.
 French : Le temps est très agréable aujourd'hui.

English: Machine learning models require large amounts of data.
 French : Les modèles d'apprentissage automatique nécessitent de grandes quantités de données.

English: How are you doing today?
 French : Comment allez-vous aujourd'hui ?

English: The meeting will start at 10 AM tomorrow.
 French : La réunion commencera demain à 10 heures.

English: Researchers are developing new methods for climate prediction.
 French : Les chercheurs mettent au point de nouvelles méthodes de prévision du climat.

English: Social media influences people's opinions.
 French : Les médias sociaux influencent les opinions des gens.

English: She completed her assignment before the deadline.
 French : Elle a terminé son affectation avant la date limite.

English: The internet connects millions of devices worldwide.
 French : Internet connecte des millions d'appareils dans le monde entier.

English: Education plays an important role in society.
 French : L'éducation joue un rôle important dans la société.

English: Natural language processing enables computers to understand human language.
 French : Le traitement du langage naturel permet aux ordinateurs de comprendre le langage humain.

English: Can you help me solve this problem?
 French : Pouvez-vous m'aider à résoudre ce problème ?

English: Technology improves communication across the world.
 French : La technologie améliore la communication à travers le monde.

English: Students are learning machine translation in this laboratory.
 French : Les étudiants apprennent la traduction automatique dans ce laboratoire.

English: The government announced a new economic policy.
 French : Le gouvernement a annoncé une nouvelle politique économique.

STEP 5 — Translate into Multiple Languages

```

print("Multi-Language Translation\n")

for sentence in sentences:
    fr = translator_en_fr(sentence)[0]['translation_text']

```

```
fr = translator_en_fr(sentence)[0]['translation_text']
de = translator_en_de(sentence)[0]['translation_text']
hi = translator_en_hi(sentence)[0]['translation_text']

print("English :", sentence)
print("French  :", fr)
print("German   :", de)
print("Hindi    :", hi)
print("-----")
```

Multi-Language Translation

English : Artificial intelligence is transforming modern technology.
 French : L'intelligence artificielle transforme la technologie moderne.
 German : Künstliche Intelligenz transformiert moderne Technologie.
 Hindi : खगोलीय विज्ञान आधुनिक तकनीक को परिवर्तित कर रहा है।

English : The weather is very pleasant today.
 French : Le temps est très agréable aujourd'hui.
 German : Das Wetter ist heute sehr angenehm.
 Hindi : मौसम आज बहुत सुखद है.

English : Machine learning models require large amounts of data.
 French : Les modèles d'apprentissage automatique nécessitent de grandes quantités de données.
 German : Maschinelle Lernmodelle erfordern große Datenmengen.
 Hindi : मशीन सीखने वाले मॉडलों के लिए बहुत मात्रा की आवश्यकता होती है।

English : How are you doing today?
 French : Comment allez-vous aujourd'hui ?
 German : Wie geht es dir heute?
 Hindi : आज आप कैसे कर रहे हैं?

English : The meeting will start at 10 AM tomorrow.
 French : La réunion commencera demain à 10 heures.
 German : Das Treffen beginnt morgen um 10.00 Uhr.
 Hindi : बैठक 10 कल 10 बजे शुरू होगा.

English : Researchers are developing new methods for climate prediction.
 French : Les chercheurs mettent au point de nouvelles méthodes de prévision du climat.
 German : Forscher entwickeln neue Methoden zur Klimavorhersage.
 Hindi : खोजकर्ताओं का कहना है कि जलवायु की भविष्यवाणी के लिए नए तरीके ईजाद हो रहे हैं ।

English : Social media influences people's opinions.
 French : Les médias sociaux influencent les opinions des gens.
 German : Soziale Medien beeinflussen die Meinungen der Menschen.
 Hindi : सामाजिक मीडिया लोगों की राय को प्रभावित करता है.

English : She completed her assignment before the deadline.
 French : Elle a terminé son affectation avant la date limite.
 German : Sie hat ihren Auftrag vor Ablauf der Frist erfüllt.
 Hindi : वह मृत रेखा से पहले अपनी नियुक्ति पूरी कर चुकी थी ।

English : The internet connects millions of devices worldwide.
 French : Internet connecte des millions d'appareils dans le monde entier.
 German : Das Internet verbindet Millionen von Geräten weltweit.
 Hindi : इंटरनेट दुनिया भर के लाखों उपकरणों से जुड़ता है.

English : Education plays an important role in society.
 French : L'éducation joue un rôle important dans la société.
 German : Bildung spielt in der Gesellschaft eine wichtige Rolle.
 Hindi : शिक्षा समाज में एक महत्वपूर्ण भूमिका अदा करती है ।

English : Natural language processing enables computers to understand human language.
 French : Le traitement du langage naturel permet aux ordinateurs de comprendre le langage humain.
 German : Natürliche Sprachverarbeitung ermöglicht Computern, menschliche Sprache zu verstehen.
 Hindi : प्राकृतिक भाषा सीखने के लिए कंप्यूटरों को इंसान की भाषा सीखने में मदद देती है ।

English : Can you help me solve this problem?

STEP 6 — Translation Quality Analysis

The translation results show that the Transformer models produce fluent and grammatically correct translations for most simple and medium-length sentences. The meaning of the original sentence is usually preserved across different languages. Technical sentences related to artificial intelligence and machine learning are also translated reasonably well, although some specialized terms remain untranslated or slightly modified.

French translations appear more natural and fluent, while German translations sometimes change word order due to grammatical rules. Hindi translations generally maintain the meaning but occasionally simplify complex phrases.

Long sentences may experience minor grammatical inconsistencies, especially when translating complex structures. Questions are translated correctly in most cases, preserving interrogative form. Overall, the pretrained models demonstrate strong translation capability, but minor inaccuracies may appear in technical or context-heavy sentences.

STEP 7 — Test with Complex Sentences

```
complex_sentences = [  
    "Although machine learning models require large datasets, they can produce highly accurate predictions.",  
    "If the weather improves tomorrow, we will conduct the experiment outside.",  
    "Researchers who study artificial intelligence are developing systems that can understand human language.",  
    "Can machine translation replace human translators completely?",  
    "The neural network model improved accuracy after several training iterations."  
]  
  
for sentence in complex_sentences:  
    print("English:", sentence)  
    print("French :", translator_en_fr(sentence)[0]['translation_text'])  
    print()
```

English: Although machine learning models require large datasets, they can produce highly accurate predictions.
French : Bien que les modèles d'apprentissage automatique nécessitent de gros ensembles de données, ils peuvent produire des pré

English: If the weather improves tomorrow, we will conduct the experiment outside.
French : Si le temps s'améliore demain, nous mènerons l'expérience à l'extérieur.

English: Researchers who study artificial intelligence are developing systems that can understand human language.
French : Les chercheurs qui étudient l'intelligence artificielle développent des systèmes capables de comprendre le langage huma

English: Can machine translation replace human translators completely?
French : La traduction automatique peut-elle remplacer complètement les traducteurs humains?

English: The neural network model improved accuracy after several training iterations.
French : Le modèle de réseau neuronal a amélioré la précision après plusieurs itérations de formation.